

For each term: write the definition from the lesson AND the analogy or memory aid the lesson gives you.

1. The lesson states that Cell Theory has THREE principles. Write each principle in full AND explain its significance:

4. The lesson mentions Robert Hooke (1665) and then Schleiden, Schwann and Virchow (1830s–1850s). There is a gap of nearly 200 years. Why do you think it took so long for Cell Theory to be formally established after Hooke first saw cells?

5. The lesson identifies THREE structures found in plant cells but NOT in animal cells. For each: name it, explain its function, and explain why animal cells do NOT need it:

6. Complete the full comparison table from the lesson. For EACH feature, tick the correct columns AND give the reason for any difference:

7. The lesson gives the magnification formula: Total magnification = eyepiece lens $\times$ objective lens. Using this formula: (a) Calculate total magnification if eyepiece = $\times 10$ and objective = $\times 40$. (b) A cell appears 3 mm on a slide at this magnification. What is its actual size in mm? Show your working. (c) Why do scientists always start with the LOWEST objective lens, not the highest?

8. The lesson explains that a wilting plant means its central vacuole is not full of water. Explain the chain of events: what happens to the vacuole when a plant is deprived of water, what then happens to the cell, and what the visible result is. Use the lesson's term "turgor pressure" in your answer.

9. The lesson describes mitochondria as "the powerhouse of the cell." Using the lesson's content, explain why muscle cells would contain MORE mitochondria than skin cells. What does this tell us about the relationship between a cell's function and its structure?
